

Defend your country - How to build your own FPV drone

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Defend your country - How to build your own FPV done

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Mark 4 frame assembling

The first step is the easiest one. You simply unfold all the items from the *Mark 4 frame package* and assemble it like a constructor in your childhood. Everything you need is just a screwdriver (M3) and the *Mark 4 7" Schematic*.



Mark 4 7" frame assembling

Professionals recommend not tightening the bolts at this stage. You will do that later, after soldering and assembling the entire drone. Simply screw the bolts softly with a hex-screwdriver M3 and enjoy the process.

EMAX motors installing

When the frame is ready, the next step is preparing and installing the motors. I used *EMAX ECOII Series ECO II 2807 6S 1300KV* and a special protective coverage for the motor wires, look at the picture below.



At this step, it is recommended to use a hex-screwdriver (M3) to install the motors without tightening the bolts. We will do so later.



Look at the picture above. This is what we should have so far before starting soldering. All 4 *EMAX ECOII* motors installed on *Mark 4 7"* drone frame. Let's continue to build our warrior device.

Power supply cable soldering

And here we are. This is perhaps the most challenging part of the job — soldering the power cables. It's beneficial if you have any experience in soldering, otherwise, I would recommend searching on YouTube and watching the video '[How to solder the power supply wires to FC](#)'.



Remember to solder the *1000uF low ESR Capacitor* included in the *Speedy Bee F405 V3 Stack package*. Connect the Red wire to red (+) and the Black wire to black (-). This is necessary to protect the ESC board.

EMAX motors soldering

Before this project, I had never had any experience in soldering, and it became a challenge for me. This is my very first attempt at soldering, as you may see from the picture below, where I soldered all four motors to the ESC board.

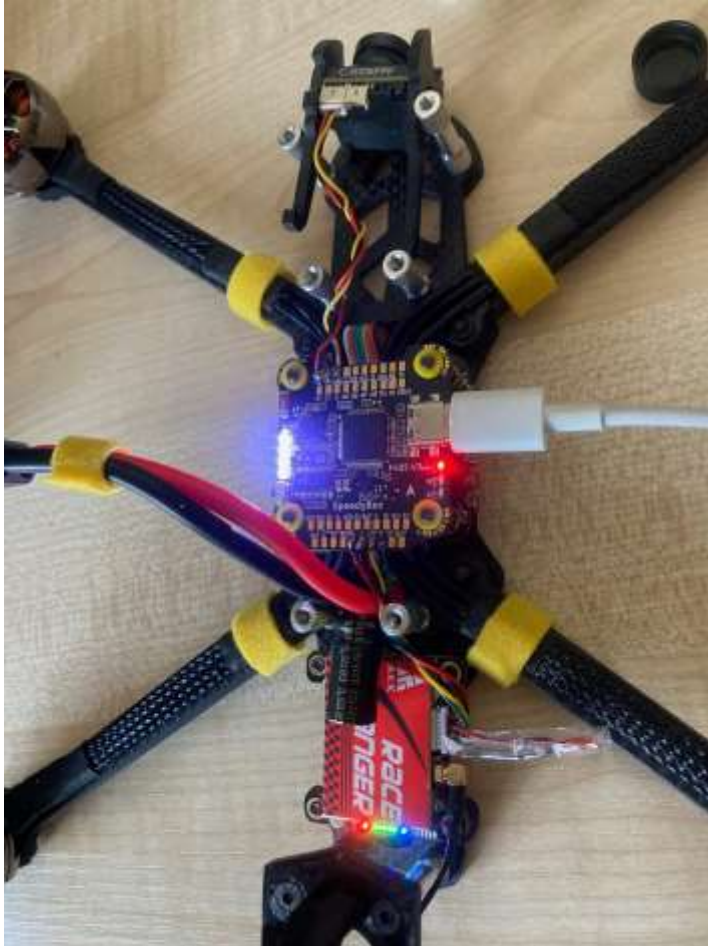


After finishing soldering the motors to the board, you need to use a multimeter to test all the contacts. Please note that contacts from different motors should not be connected or show continuity.

At this stage, you can already connect the ESC to the FC with an 8-pin cable and link the entire device to a computer with [Betaflight Configurator](#) (using a USB cable) to pre-test the motors. However, I will explain this step later on in the section '*Betaflight configuration*'.

VTX and camera setup

According to the wiring diagram on the official website of [Speedy Bee](#), the *Video Transmitter VTX AKK Race Ranger 1.6W* should be connected to the **9V, G, VTX, and T1** pins on the first UART port of the FC. T1 means that this is a TX from *UART1*. Look at the picture above.



Always double-check all the pins when connecting any peripheral device to the flight controller because they may vary from device to device. For the VTX, we use **red (1) and black (2)** wires, as well as **yellow (5) and green (6)**. The internal **red (3) and black (4)** wires should not be used. I have isolated them. These may be used to connect to the video camera directly, but we will be using the FC to connect with the camera through that.

Video-camera *Caddx Ratel 2 12000 TVL* connects to the flight controller using **5V, G, and CAM** pins as shown on the picture above. **Red (1)** to 5V, **Black (2)** to G, and **Yellow (3)** to CAM.

After soldering all the necessary connections, you can proceed to connect the FPV system either to your computer or directly to the battery for testing. If there is no smoke, congratulations, you have completed the work correctly. From this point, you can even turn on your *FPV glasses* and search for the channel to search your *FPV drone*. It should already be operational by this point if **UART1** is configured correctly. [Read later on].

IMPORTANT!

Don't forget to connect your FPV antenna, Lollipop 4 V4 SMA 10cm, if you power on the VTX. Otherwise, there is a chance it could burn down.

ELRS receiver setup

The ELRS receiver is used to connect your drone with the Radio Controller. The operator manages the drone's direction, speed, turns, altitude, and everything else using this radio controller. It is the primary device for controlling the drone.



According to the official documentation, it should be connected to **UART2** on the FC board. Double-check which one you use and connect it according to the specification. In my case, as I am using *Express LRS Nano*, so that I connected **Black to G**, **Red to 4V5**, **White to R2**, and **Yellow to T2**.

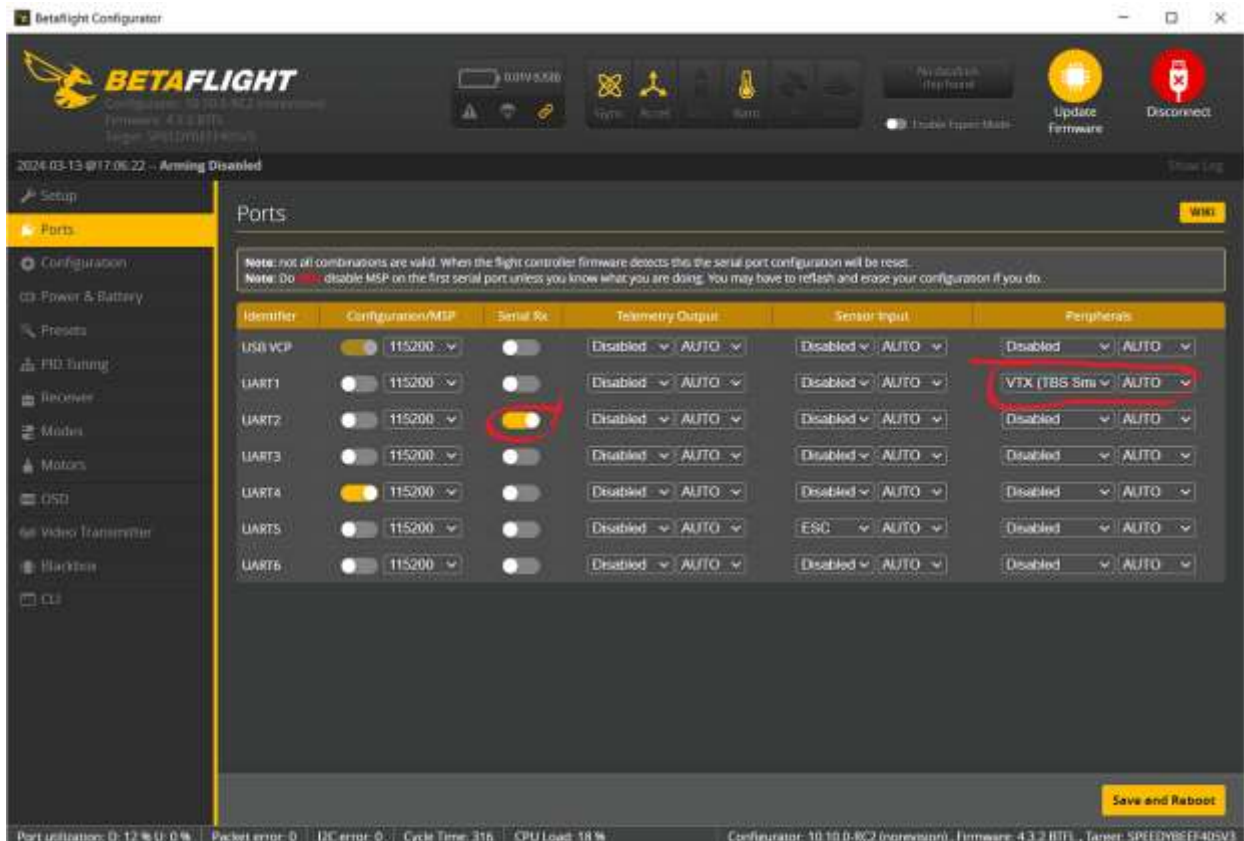
But before that, I also soldered these wires to the ELRS receiver exactly and double-checked that TX from the receiver should be connected with RX (R2) on the controller. At the same time, RX on the receiver should be connected to TX (T2) on the controller.

Once you have done this, you can power on your FC and pair the device with your *RadioMaster TX12* or analog radio controller to check its functionality and readiness.

Software Part

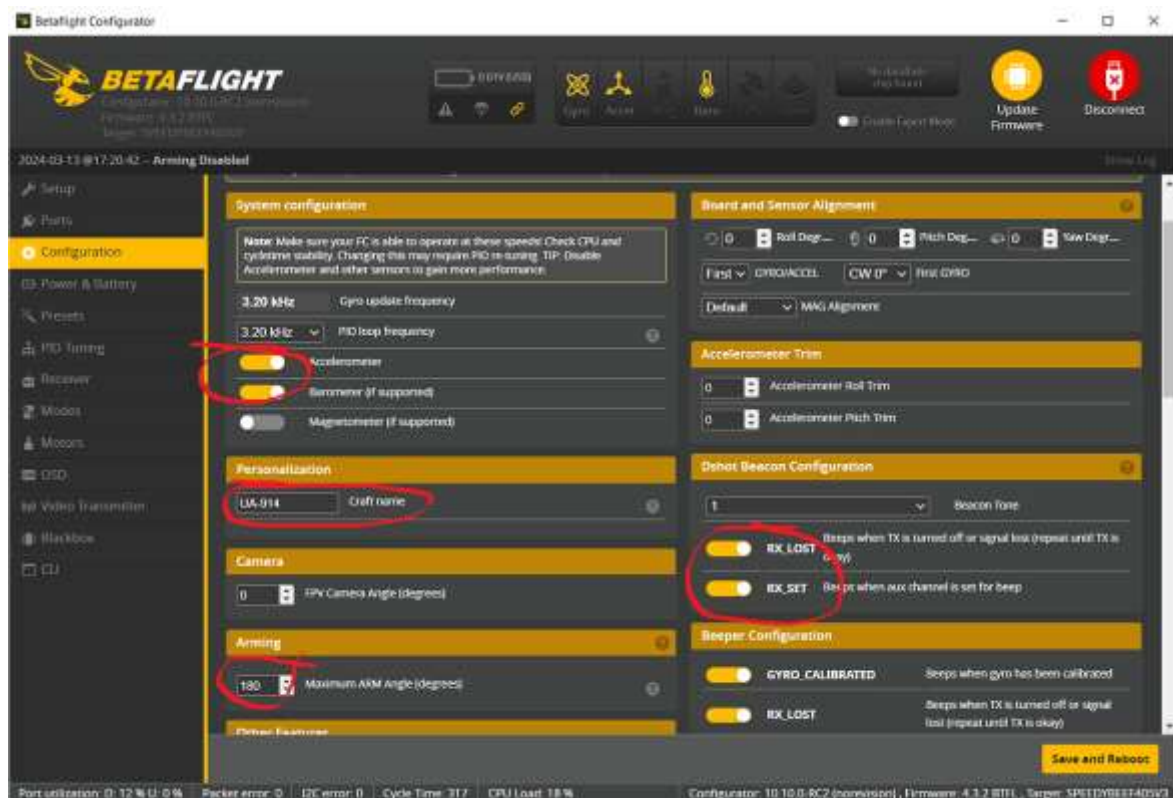
Betaflight Configuration

Alright, we've completed the soldering, so we're good to move on. Next up, let's configure the parameters for our Drone. To do that, you'll need to download the [Betaflight Configurator](#) for your operating system and connect your drone to the computer via USB.



On the first screen you will likely see the image of your drone and some general information. We skip this page and go to the 'Ports' tab to properly check and setup parameters.

Since we've soldered the VTX on **UART1**, we need to select one of the VTX-related types from the list under the 'Peripherals' section. In my scenario, since the *Race Ranger VTX* utilizes the *SmartAudio* protocol, I've opted for 'VTX (TBS SmartAudio)'. In your case just try one of the VTX here.



The next page is the 'Configuration', where we need to enable the following:

- Enable 'Accelerometer' and 'Barometer';
- Set up your *craft's name* instead of 'UA-914' displayed on my screen;
- Enable arming even under angle (180 degrees);
- Enable 'RX_LOST' and 'RX_SET' for audible alerts.

The screenshot shows the Betaflight Configurator interface. The top bar includes the Betaflight logo, version 4.3.2, and various status icons. The left sidebar lists navigation options like Setup, Ports, Configuration, Power & Battery, Presets, and PID Tuning. The main area is titled 'PID Profile Settings' and contains several sections:

- PID Profile Settings:** A table with columns for Proportional, Integral, D Max, Derivative, and Feedforward. The values are: Proportional (50, 55, 45), Integral (40, 45, 35), D Max (40, 40, 0), Derivative (25, 26, 0), and Feedforward (90, 90, 90).
- Basic Mode:** A section with a dropdown menu set to 'OFF' and a table of settings for Damaging, Tracking, and Stick Response.
- PID Controller Settings:** A section with a dropdown menu set to 'PID' and a table of settings for Feed, Forward, Feedforward, Feedforward, Feedforward, and Feedforward.
- Throttle and Motor Settings:** A section with a dropdown menu set to 'Throttle' and a table of settings for Throttle Boost and Motor Output Limit.

Red circles highlight the PID Profile settings and the PID Controller settings. The Motor Output Limit is also highlighted with a red circle.

- Minimum cell voltage: 3.0
- Maximum cell voltage: 4.2
- Warning cell voltage: 3.1

These settings are ensuring correct operation with the power supply.

PID (Proportional-Integral-Derivative) profiles determine how the flight controller responds to various inputs and disturbances during your flight.

Betaflight Configurator

2024-03-13 @ 17:38:42 - Arming Disabled

PID Profile Settings

Basic/Name	Proportional	Integral	D Max	Derivative	Feedforward
ROLL	58	40	40	25	90
PITCH	55	45	48	28	90
YAW	45	35	0	0	90

PID Converter Settings

- Feed forward: 10, 40, OFF, 15, 90, 0.00
- gear reduction: 10
- Smoothness: 40
- Averaging: OFF
- Boost: 15
- Max Rate Limit: 90
- Threshold: 0.00

Mode

Mode: OFF (Low, Default, High)

Damping: 1 (Low, Default, High)

Tracking: 1 (Low, Default, High)

Stick Response: 1 (Low, Default, High)

Throttle and Motor Settings

- Throttle Boost: 5
- Motor Output Limit: 80

Notes:

- Note: Slider access and range is restricted because you are not in expert mode. Basic mode should be suitable for most builds and beginners.
- Note: Slider(s) are disabled because current values are outside the Basic Mode adjustment range. Switch to Expert Mode to make changes.

Angle/Rate Limit

Strength	Transition
Angle	50
Horizon	50
Angle Limit	75

Refresh Save

Port utilization: 0.12% / 0.0% | Packet error: 0 | I2C error: 0 | Cycle Time: 316 | CPU Load: 18% | Configurator: 10.10.0 RC2 (Innovations), Firmware: 4.3.2.BT1, Target: SPEEDY98ESP405V3

By adjusting these PID profile settings, you can optimize the flight performance of the drone, achieving better stability, maneuverability, and responsiveness.

I recommend configuring all the settings *highlighted in red* on the screen above to optimize performance. If you need any explanation about that, you can hover over the *question button* in front of each parameter.

The screenshot shows the Betaflight Configurator interface for PID Tuning. The left sidebar contains navigation options: Setup, Ports, Configuration, Power & Battery, Profiles, PID Tuning (selected), Receiver, Modes, Motors, OSD, Video Transmitter, Blackbox, and CLI. The main area is titled 'PID Tuning' and includes buttons for 'Copy profile', 'Copy rateprofile', 'Reset this Profile', and 'Show all PIDs'. Below these are tabs for 'PID Profile Settings', 'Rateprofile Settings' (selected), and 'Filter Settings'. The 'Rateprofile Settings' tab shows a table of 'Basic/Acro Rates' with columns for 'Center Sensitivity', 'Max Rate', 'Expo', and 'Max Vel (ang/s)'. The 'Max Rate' column is highlighted in red. The 'Rates Preview' graph shows a red curve. The 'Throttle Curve Preview' graph shows a yellow curve. The bottom status bar displays various system metrics.

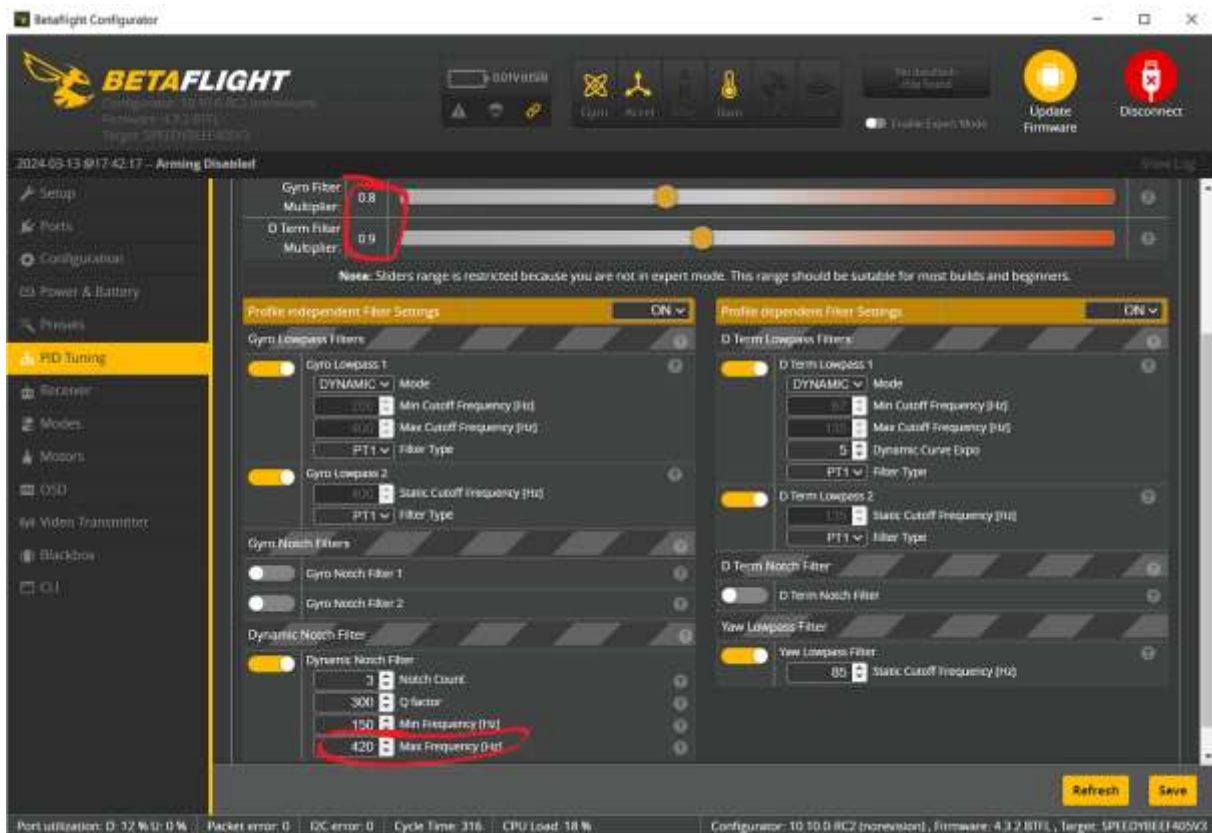
	Center Sensitivity	Max Rate	Expo	Max Vel (ang/s)
ROLL	70	490	0.00	490
PITCH	70	490	0.00	490
YAW	70	490	0.00	490

Rates Preview: 0 deg/s, 0 deg/s, 0 deg/s, 490 deg/s, 490 deg/s, 490 deg/s

Throttle Curve Preview: 0% = 0%

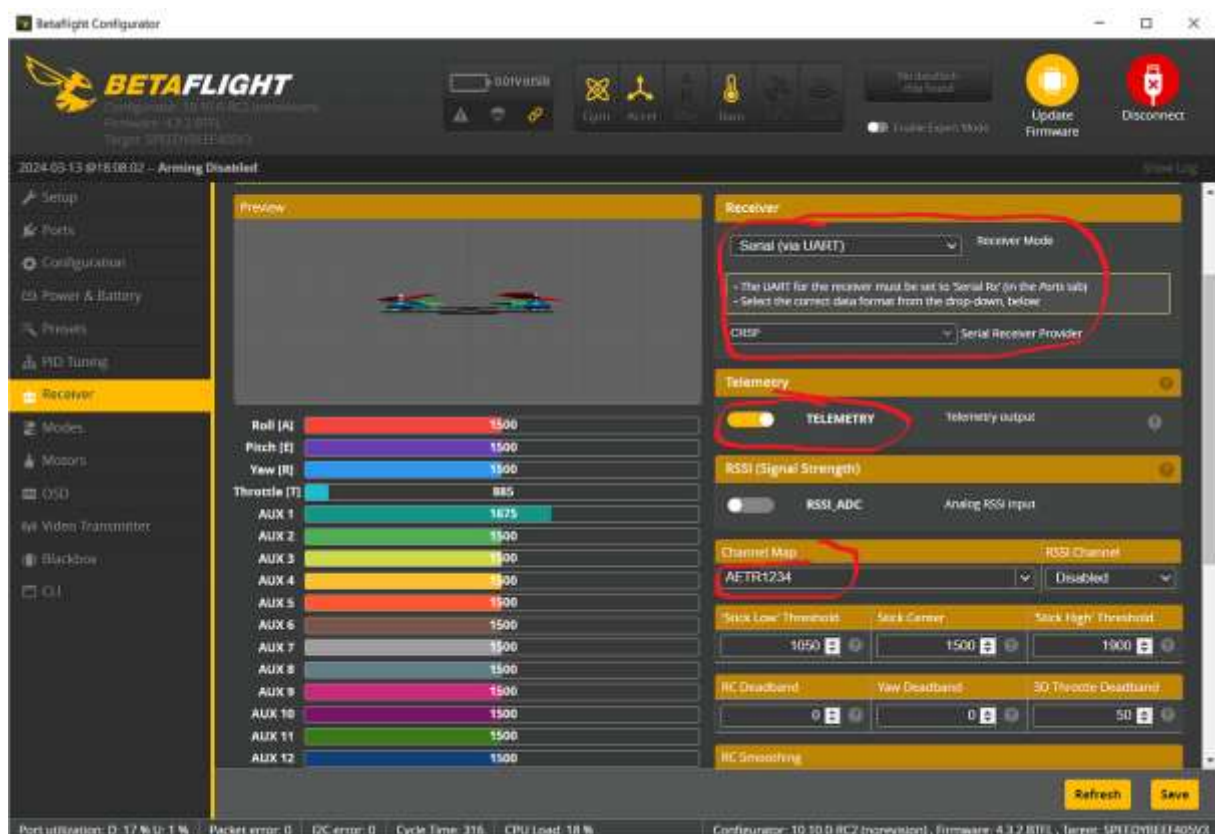
Port utilization: 0.11 % U: 0.0 % / Packet error: 0 / QX error: 0 / Cycle Time: 316 / CPU Load: 18 % / Configurator: 10.10.0 RC2 (development) / Firmware: 4.3.2 BFL / Target: SPIFYBEE-1405V3

'Rate Profile Settings' control how sensitive and responsive the drone is to stick movements, particularly regarding its rate of rotation or change in direction. These settings essentially dictate how quickly the drone reacts to pilot inputs. The default value for parameters *highlighted in red* is 620, but let's set it to 490 to increase performance.



By adjusting 'Filter Settings', you can effectively reduce noise and vibrations in the sensor data, resulting in smoother flight performance, improved stability, and better control responsiveness.

I suggest changing only three parameters *highlighted in red*. Remember to *save changes* on each page to apply them. Otherwise, they will not take effect.

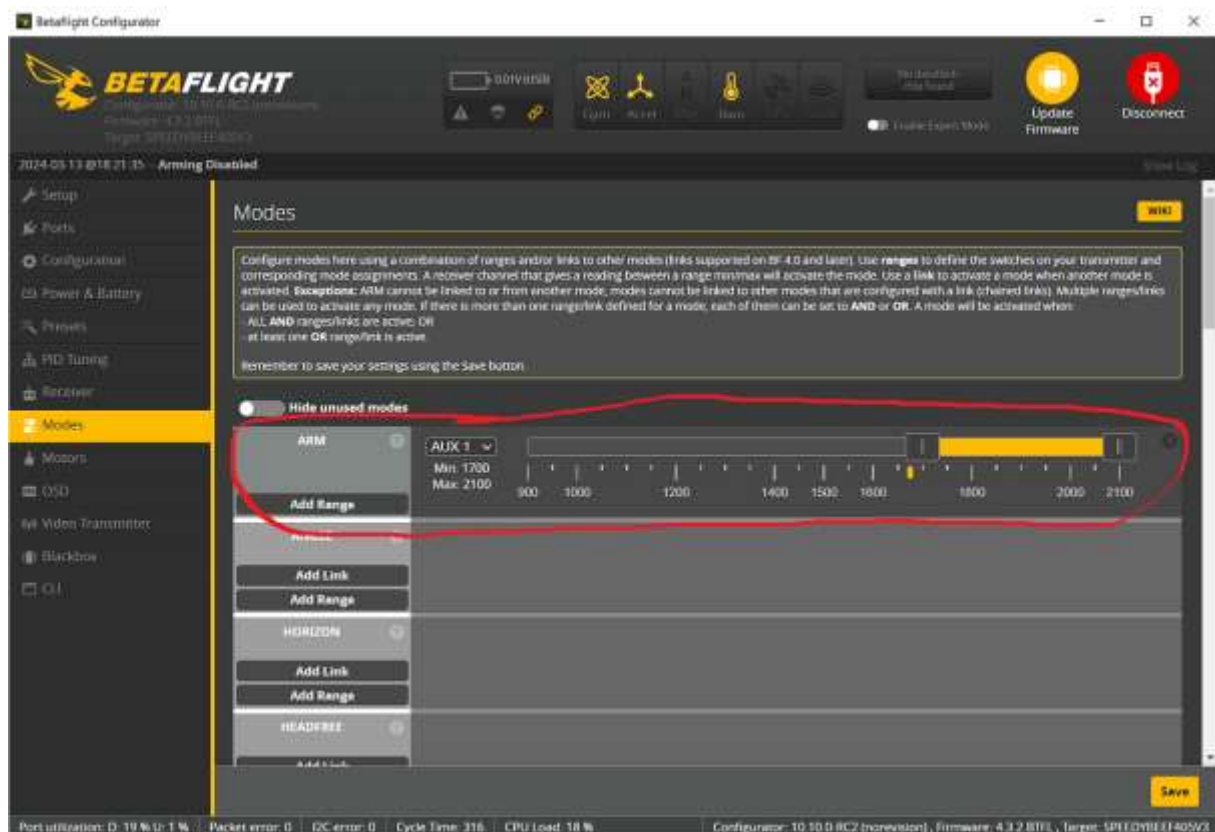


The 'Receiver' section is used to configure how the flight controller communicates with the receiver unit, in our case *ELRS 915 MHz Nano*, which receives commands from the *RadioMaster TX12* radio controller.

Since we've soldered our receiver to **UART2** as configured in the 'Ports' section above, we only need to adjust a few parameters on this page:

- Receiver mode: *Serial via UART*;
- Serial Receiver Provider: *CRSF*;
- Enable '*Telemetry*' to display key data on the radio controller;
- Channel Map: *AETR1234*.

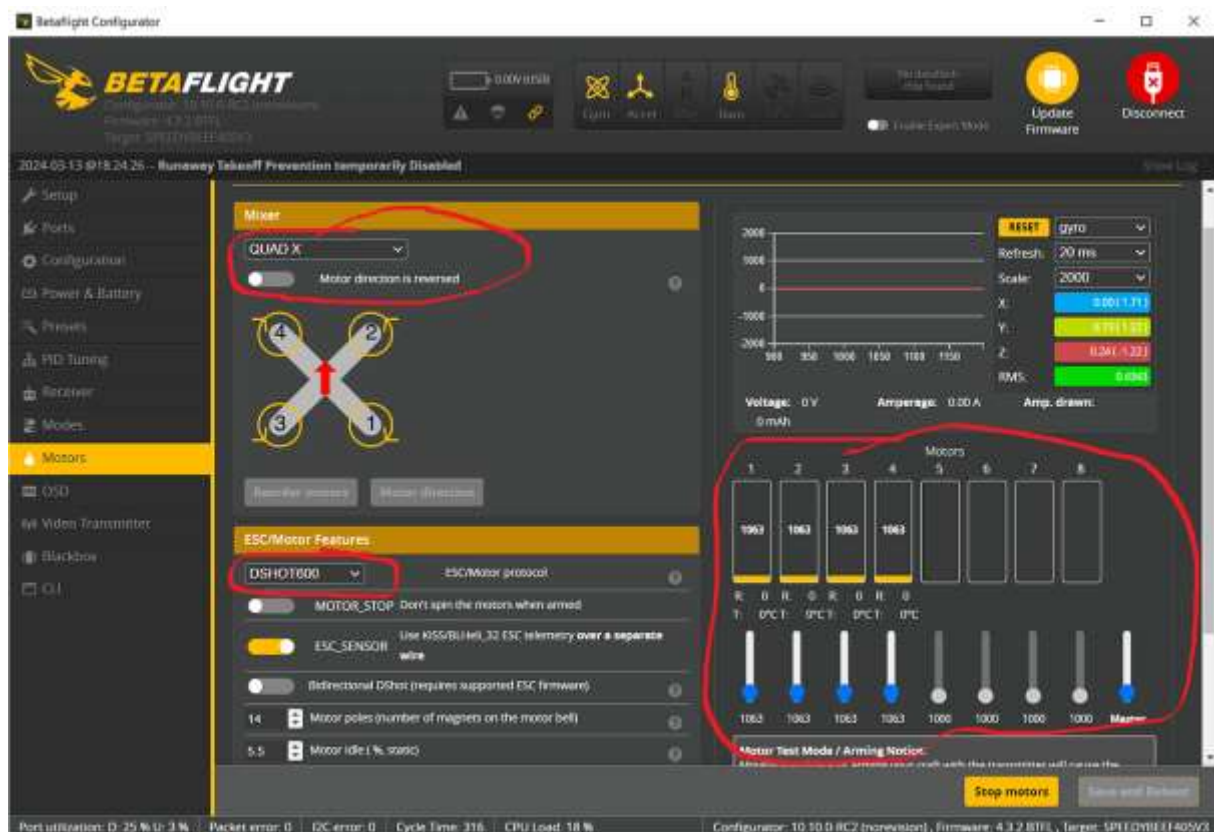
Please note that when setting up the *RadioMaster TX12*, choose the *CRSF* protocol, which we have configured in the flight controller.



Adding ARM mode in **Modes** section

Adding an 'ARM' mode in the 'Modes' section allows you to control when the drone is armed or disarmed. To set this up, you need to bind your *RadioMaster TX12 Radio Controller* with the drone during configuration. Then, click on 'Add Link' next to the 'ARM' command, and toggle the switch on/off on the *Radio Controller* that you want to use for **ARM/DISARM**. This action configures it accordingly.

Also, using the *yellow scroll*, as shown on the picture above, you can specify the tumbler state in which the drone will be armed. And don't forget to save changes in the right bottom corner of the configuration page.

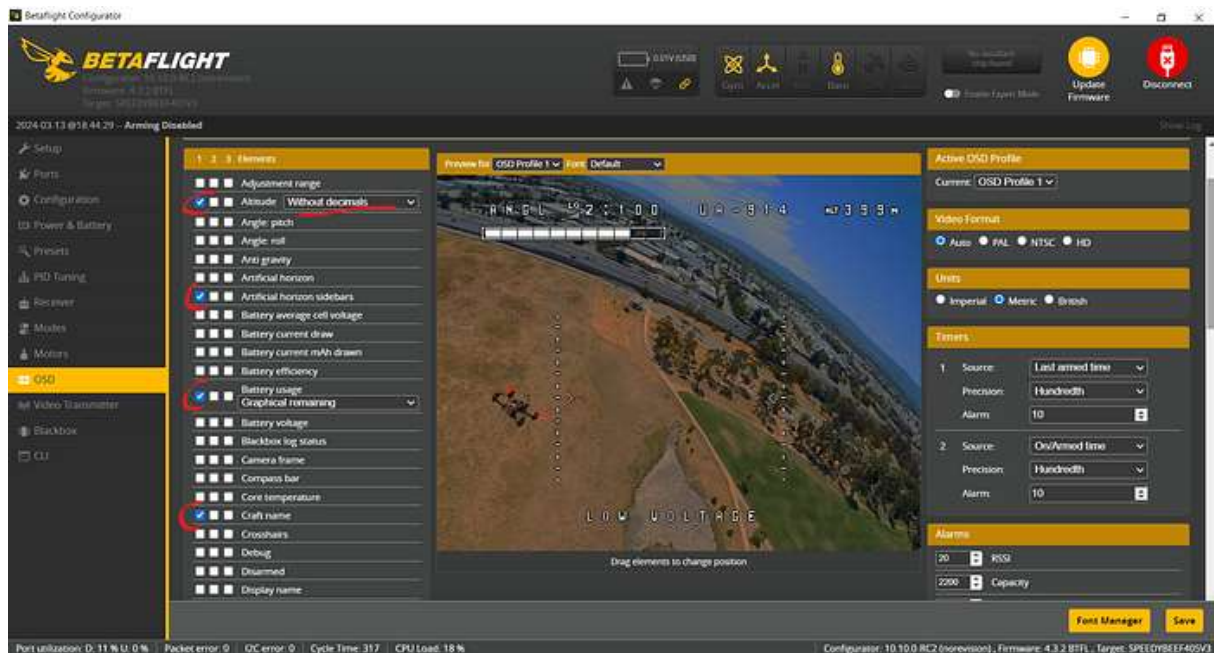


The 'Motors' section has tools for testing and resolving issues with the motors on your drone. It enables you to individually activate each motor to confirm their proper functionality and configuration.

For our setup, we should select the following options:

- Mixer: *QUAD X*;
- Motor protocol: *DSHOT600*.

As a part of motor testing, it permits you to confirm the functionality of each motor, adjust their output as needed, and ensure they spin in the correct direction for optimal flight control. Remember to utilize this feature safely by conducting tests *without propellers installed* on your drone to prevent any potential damage.



The OSD (*On-Screen Display*) section allows you to configure and customize the on-screen display settings for your FPV (First Person View) video feed displayed on your video monitor or goggles.

I would recommend to setup the following:

- Altitude (without decimals);
- Artificial horizon sidebars;
- Battery usage (graphical remaining);
- Craft name;
- Flight mode;
- Link quality;
- Warnings.

In addition to displaying various parameters on the screen, you can also drag and drop them to the specific location on the screen.

Overall it provides essential flight information at a glance, enhancing your situational awareness and overall flying experience. Properly configuring the OSD can help you monitor important telemetry data and make informed decisions while piloting the drone.

Betaflight Configurator

2024-03-13 @ 15:41:29 - Arming Disabled

Selected Mode

- ☐ Enter frequency directly
- RACEBAND** Band
- Channel 1** Channel
- 200** Power
- ☐ Pt Mode
- 0** Pt Mode frequency
- On** Low Power Disarm

Current Values

- Device ready: **False**
- VTX Type: SmartAudio Unknown
- Band: RACEBAND
- Channel: 1
- Frequency: 5658
- Power: 200
- Pt Mode: No
- Pt Mode frequency: 0
- Low Power Disarm: On

VTX Table

5 Number of bands 8 Number of channels by band

Name	Letter	Factory	1	2	3	4	5	6	7	8	
BOSCAM_A	A	☐	5885	5845	5825	5805	5785	5765	5745	5725	Band 1
BOSCAM_B	B	☐	5733	5752	5771	5790	5809	5828	5847	5866	Band 2
BOSCAM_E	E	☐	5705	5685	5665	5645	5625	5605	5585	5565	Band 3
FATSHARK	F	☐	5740	5790	5780	5800	5820	5840	5860	5880	Band 4
RACEBAND	R	☐	5668	5695	5732	5769	5806	5843	5880	5917	Band 5

4 Number of power levels

0 1 2 3 4 value

Save Lua Script Save to file Load from file Load from clipboard Save

Port utilization: 0: 11 % U: 0 % Packet error: 0 CRC error: 0 Cycle Time: 316 CPU Load: 18 % Configurator: 10.10.0-RC2 (development) Firmware: 4.3.2 BFL Target: SPI12YB-E1405V3

The VTX is responsible for transmitting the video feed from your drone's onboard camera to your FPV (First Person View) goggles or video receiver. As you remember we have soldered our video transmitter to UART1 on FC.

Our *AKK Race Ranger 1.6w* has a specific VTX table which you have to download from the file.
Look at the example below:

```
{
  "description": "AKK Race Ranger VTX Config file",
  "version": "1.0",
  "vtx_table": {
    "bands_list": [
      {
        "name": "BOSCAM_A",
        "letter": "A",
        "is_factory_band": true,
        "frequencies": [
          5865,
          5845,
          5825,
          5805,
          5785,
          5765,
          5745,
          5725
        ]
      },
      {
        "name": "BOSCAM_B",
        "letter": "B",
        "is_factory_band": true,
        "frequencies": [
          5733,
          5752,
          5771,
          5790,
          5809,
          5828,
          5847,
          5866
        ]
      },
      {
        "name": "BOSCAM_E",
        "letter": "E",
        "is_factory_band": true,
        "frequencies": [
          5705,
          5685,
          5665,
          5645,
          5885,
          5905,
          5925,
          5945
        ]
      },
      {
        "name": "FATSHARK",
        "letter": "F",
        "is_factory_band": true,
        "frequencies": [
          5740,
```

```
        5760,
        5780,
        5800,
        5820,
        5840,
        5860,
        5880
    ]
},
{
    "name": "RACEBAND",
    "letter": "R",
    "is_factory_band": true,
    "frequencies": [
        5658,
        5695,
        5732,
        5769,
        5806,
        5843,
        5880,
        5917
    ]
}
],
"powerlevels_list": [
    {
        "value": 0,
        "label": "200"
    },
    {
        "value": 1,
        "label": "400"
    },
    {
        "value": 2,
        "label": "800"
    },
    {
        "value": 3,
        "label": "1600"
    }
]
}
}
```

If you have any other VTX, you should find an appropriate VTX table and load it in this section by clicking on the 'Load from file' button. Once you're done with this, you can proceed to choose the following settings:

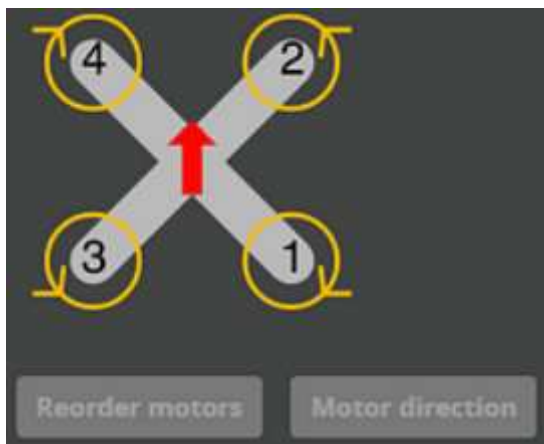
- Band: RACEBAND;
- Channel: 1;
- Power: 200.

Also, enable 'Low power disarm' to conserve energy when the aircraft is disarmed and stay on the ground surface. Each time you do any changes don't forget to save them clicking 'Save' button usually located in the right bottom corner of the configuration section page.

Propellers installation

When you are done with all the configuration in **Betaflight**, you are ready to tighten bolts, install the battery onboard, and attach the propellers. Installing propellers is crucial for a successful flight. Installing them in the wrong direction can cause the drone to flip or sustain damage on its first flight. So, be careful.

In our configuration, we use the **direct mode** for motors. This means that the *front left* and *back right* motors rotate to the right side as [clockwise]. The *right front* and *left bottom* motors rotate [counterclockwise] to the left side. This is shown in the picture below.



Each blade on the propeller has two sides: the side forming the top and the one at the bottom. The propeller should be installed on the motor so that the **upper part of the blade faces the direction of motor rotation.**

As shown on the picture below, I am tightening nut on the *left bottom motor (3)*, which rotates *counterclockwise*. As you can see, upper part of the blade faces the left direction.



Tighten the nuts securely on all motors, but be careful not to over-tighten, as this can damage the motor or propeller. When done, ensure that everything is installed correctly, open the camera, bind with Radio Controller and Goggles, and you are ready for the first flight.

Drop system installation

Disclaimer:

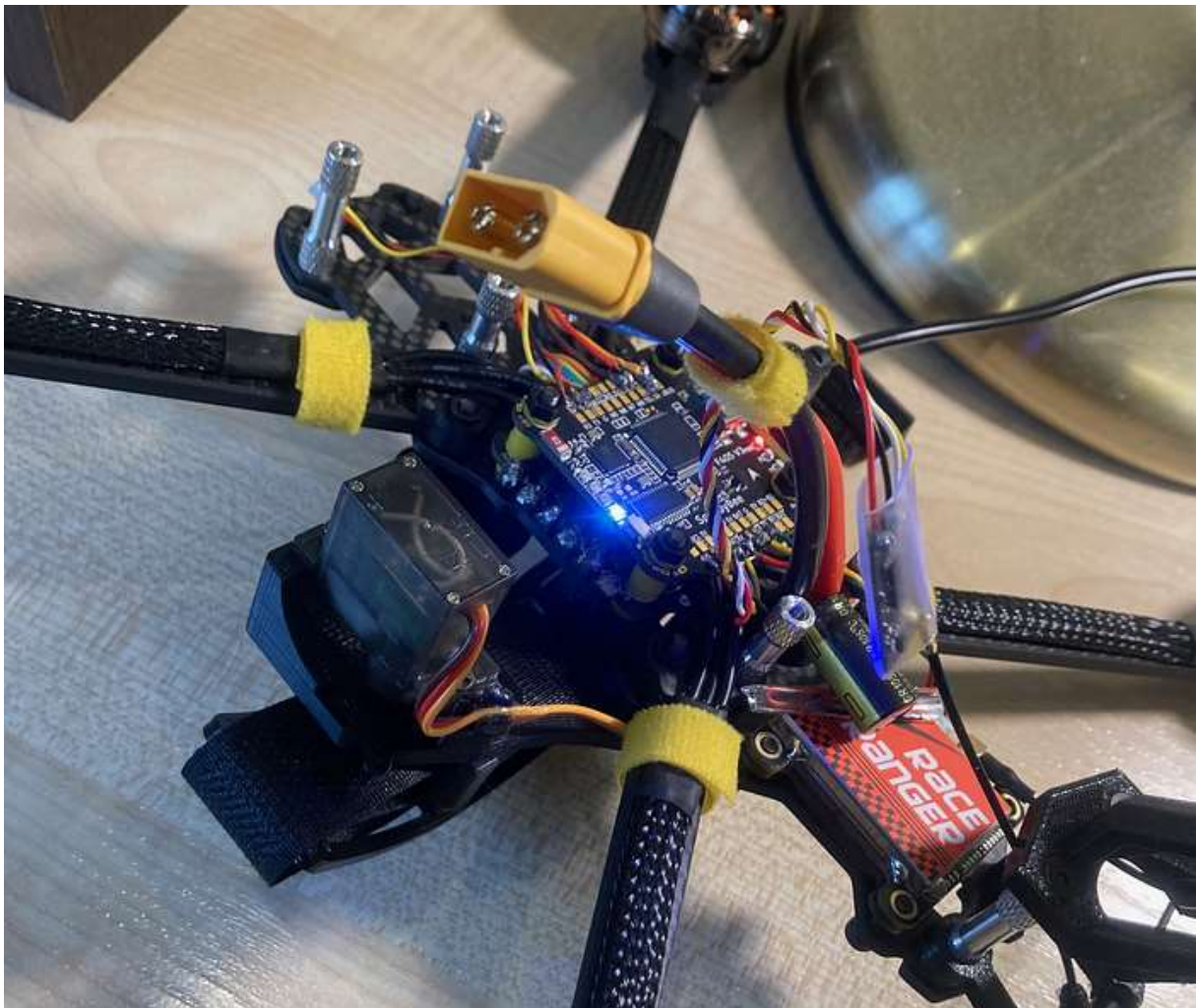
I won't release the CAD-files for the payload system as of now, it's not mine to release.

I will do my own design and release them.

They will however follow the same logic as below.

The *Drop System* is the final step of configuration which transforms our standard drone into a **military-grade tool**, enabling the deployment of bombs. From a technical perspective, it includes a servo mechanism capable of holding and releasing bombs when the operator switches the toggle to the bombing position.

Press enter or click to view image in full size

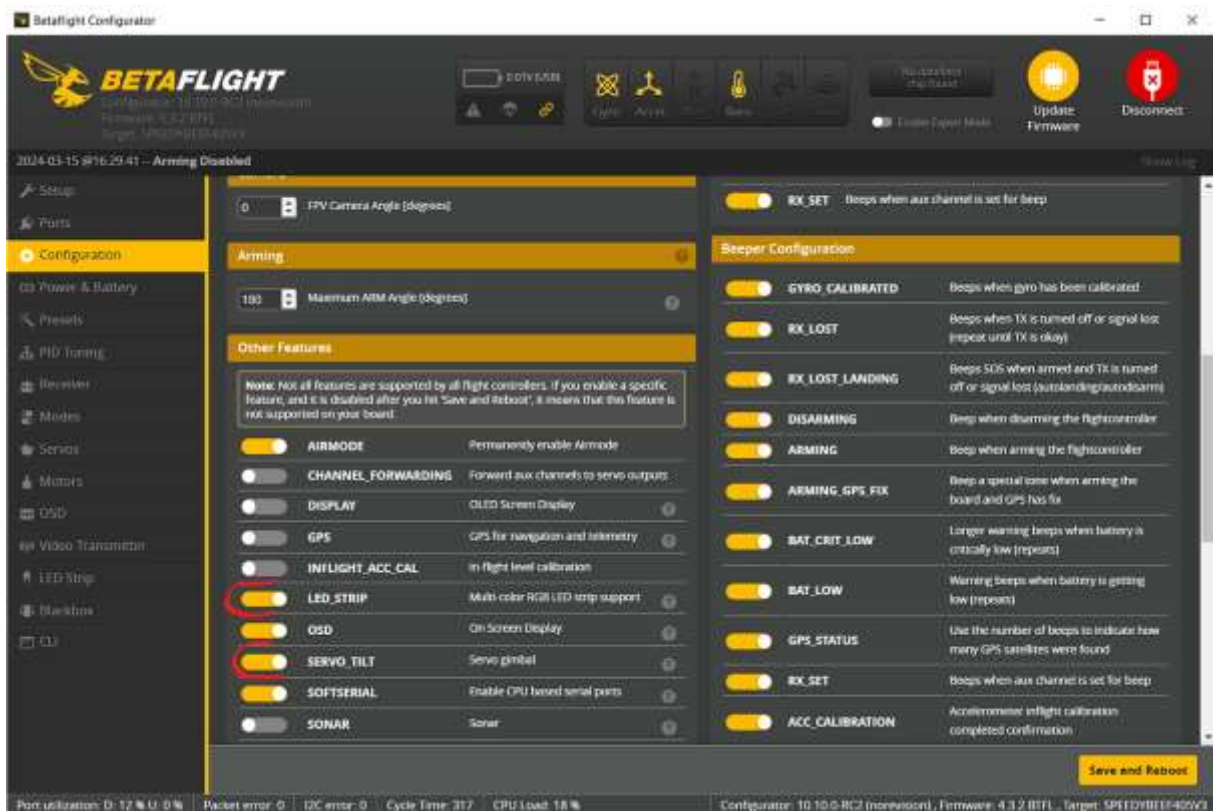


Servo module installed on Drone

I've connected the servo mechanism to the flight controller as follows:

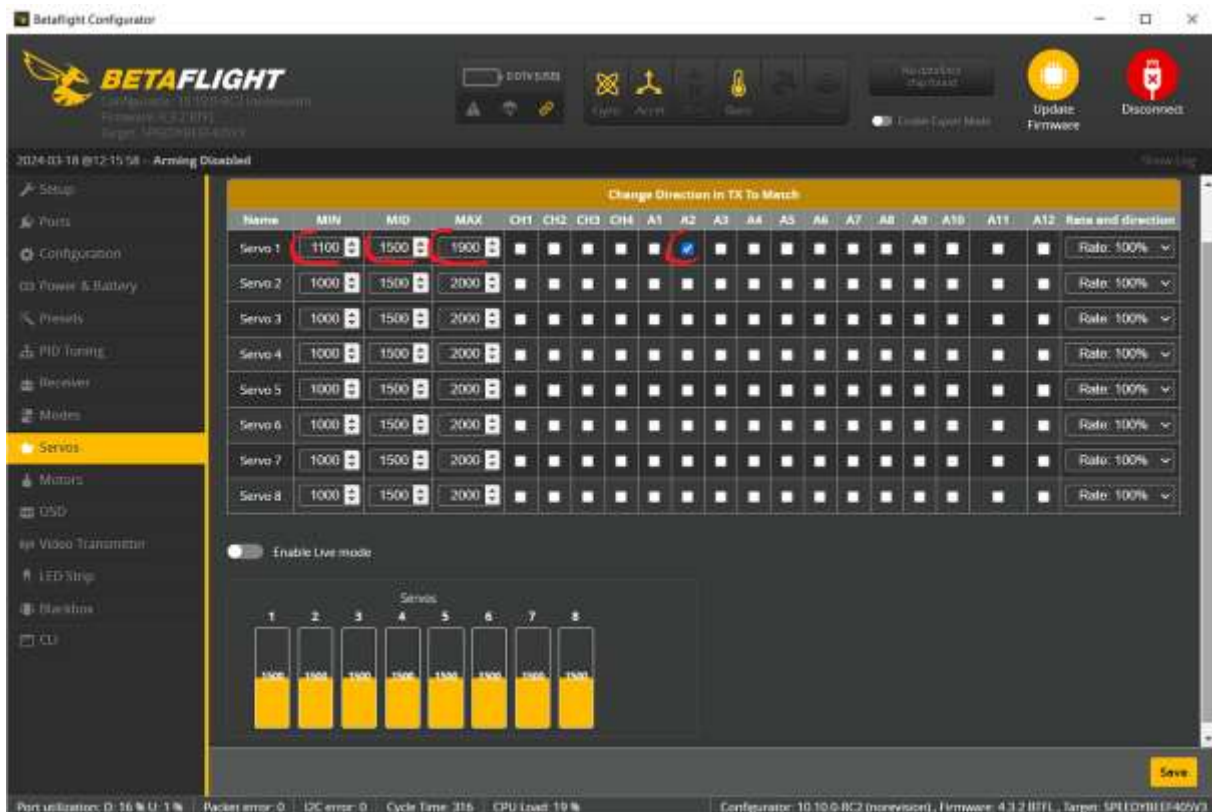
- **Red** wire of servo is soldered to the **5V pin** on the top right corner of FC;
- **Black** wire of servo is soldered to the **G pin** on the top right corner of FC;
- **Yellow** wire of servo is soldered to the **S9 pin** on the FC.

Once you done with soldering you have to configure these last things in the *Betaflight Configurator* as shown below.



Enabling **SERVO_TILT** and **LED_STRIP**

As we are using the **5V** and **G** pins from the LED strip of the FC, we also need to enable them in the configuration by enabling *LED_STRIP*. Additionally, enabling *SERVO_TILT* allows us to configure the **servo mechanism** on one of the toggles.



Servo configuration

As shown on the picture above, I have selected *mid* (1500), *min* (1100), and *max* (1900) values for Servo 1, determining its angle of rotation. Additionally, I've configured A2 (AUX2 toggle) to serve as the bombing function.

Remember to save changes in all sections once you've finished configuration. Now, disconnect your drone from the computer, connect it to the battery, insert a *tennis ball* into the servo mechanism instead of a bomb, and test it using your *RadioMaster TX12* radio controller.

Ready for combat operations

As I mentioned in the beginning of this story, a lot of my friends took part in this project by purchasing different components for drone and drone kit. Some of them helped me with soldering, as it's my very rare attempt to solder something. And finally, we assembled the comprehensive **Combat Unit** comprising all the necessary elements for fighting with enemies.



FPV Combat Unit

As you may see from the picture above, the *FPV Drone Kit* is now ready for combat operations. It includes: 7" Drone, Li-Ion battery 9000 mAh, 5.8G FPV Goggles, RadioMaster TX12 Radio Controller, ELRS 915 MHz adapter, and Drop system attached with a *tennis ball instead of a bomb installed*. The operator will change it later, on the frontline.

As the final part of the project, this equipment was handed over to the Ukrainian military, who heroically defended the eastern border of the country. It will reduce risks of injury for soldiers involved in combat missions on the frontline, thereby attaining tactical superiority.

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